Project Design Phase-I
Proposed Solution Template

Date	1 March 2026
Team ID	
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Plant diseases cause severe crop damage, leading to massive financial losses for farmers and threatening global food security. Identifying these diseases manually is time-consuming, prone to human error, and requires scarce agricultural experts. Late or incorrect diagnosis results in delayed treatment, decreased crop yields, and frustration for both commercial farmers and home gardeners.
2.	Idea / Solution description	PlantCare AI is an automated plant disease detection system powered by Deep Learning. Users upload images of plant leaves via a web or mobile interface (built with Flask). The system uses a highly efficient, pre-trained MobileNetV2 Convolutional Neural Network (CNN) via transfer learning to instantly analyze the image and classify it into one of 38 distinct disease classes, providing real-time, accurate diagnostic feedback.
3.	Novelty / Uniqueness	The core uniqueness lies in its highly optimized architecture. By leveraging Transfer Learning with MobileNetV2, PlantCare AI achieves state-of-the-art accuracy while keeping computational costs and processing time extremely low. Furthermore, it is a unified solution adaptable to three distinct environments: automated IoT farm monitoring, a consumer mobile app, and an interactive EdTech tool for agricultural training.
4.	Social Impact / Customer Satisfaction	**For Farmers:** Maximizes crop yields and prevents financial ruin by enabling early, precise

		intervention. It also prevents the overuse of harmful pesticides through targeted treatment. **For Gardeners:** Democratizes expert-level plant care, bringing peace of mind and healthy gardens to everyday hobbyists. **For Students:** Provides an interactive, hands-on learning tool that modernizes agricultural education.
5.	Business Model (Revenue Model)	• **Freemium App (B2C):** Free basic disease identification for home gardeners, with a premium subscription for advanced care plans and garden tracking. • **Enterprise Subscriptions (B2B):** Paid API access and integration services for large-scale farms integrating PlantCare AI into their automated drone or IoT monitoring systems. • **EdTech Licensing (B2B):** Licensing the software to agricultural universities and training programs.
6.	Scalability of the Solution	The system is intrinsically highly scalable. The lightweight MobileNetV2 model allows for seamless deployment on mobile edge devices, local IoT networks (like Raspberry Pi for offline farm use), and centralized cloud servers. The Flask web backend can handle massive concurrent user requests when deployed on cloud infrastructure (like AWS or Google Cloud). Additionally, the model can be easily updated and scaled to detect hundreds of new diseases simply by adding new images to the training pipeline.